

Mackenzie Goodwin

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SKILLS

Languages: Python, C, C++, SystemVerilog, Verilog, Java, JavaScript, Solidity

Tools: Verilator, Cocotb, Altium, Vivado, Cadence, Ansys HFSS, ModelSim

Experience: Network Protocols, RTL Design, High-Speed Design, Full Stack, Analog & Digital Design, PCB Design, Web Design

EXPERIENCE

Tesla Inc.

Palo Alto, CA

Staff Silicon Design Engineer

Aug 2025 – Present

- Building next-generation DOJO D3 silicon design, focusing on scale-up and scale-out domain architecture
- Explored scale-up options including NVLink, THU, Google's Torus, and AMD's RoCE
- Own boot architecture for AI6 across boot sequence, NoC, CPU, Ethernet, UFS, and related subsystems
- Taped out two chips, brought up systems, and validated designs; prototyped critical paths in FPGA and ran production workloads until silicon returned
- Worked with Samsung and TSMC foundries on dual-foundry tape-out, developing process-specific ECOs and parallelizing boot and related block delivery to support a 9-month design cycle

Tesla Inc.

Palo Alto, CA

Senior Silicon Design Engineer

Feb 2024 – Aug 2025

- Developed the proprietary Tesla Transport Protocol (TTP) for low-latency, high-throughput communication across DOJO's die-to-die fabric in SystemVerilog RTL
- Advanced TTP from design phase to tape-out readiness in SystemVerilog on next generation DOJO Z2
- Designed TTP RDMA systems incorporating Scale Up Ethernet (SUE) standard + L3/L4 protocol and routing
- Scaled data center for video clip tensorization ingest and all-reduce operations

Tesla Inc.

Palo Alto, CA

Hardware Engineering

Jul 2022 – Feb 2024

- Led custom compute cluster design integrating TTP-enabled NICs for die-to-die fabric communication
- Brought DOJO interface components to production with Node.js diagnostic tools and signal integrity debugging
- Wrote RTL enabling microcontroller access to internal buses and hostless hardware control
- Designed hardware and manufacturing processes for high-power AC to DC conversion on DOJO cabinet power rails

Tesla Inc.

Palo Alto, CA

DOJO Hardware Intern

Aug 2021 – Dec 2021

- Tested high-speed mesh network interfaces achieving TB/s throughput
- Engineered supercapacitor hot-swap controller for live cabinet insertion
- Integrated complex DOJO cabinet systems with harnessing and power delivery

Tesla Inc.

Palo Alto, CA

Autopilot Hardware Intern

Jan 2021 – May 2021

- Designed PCB schematics and layouts for Autopilot hardware subsystems and brought up multiple board systems including switch, GPS, and VRM
- Created hardware validation testing suites in Python for Ethernet and GPS components
- Resolved reliability issues across temperature-varying environments
- Performed signal integrity analysis on SGMII and 1000Base-T1 paths

Kazoo Technology

Hong Kong, HK

Electronics Designer Intern

Aug 2019 – Apr 2020

- Reverse engineered a proprietary stylus touch system and built analog front-end amplifiers so the hardware could mimic Apple Pencil-style touch
- Built an FPGA touch-emulation setup with a 200 MSPS ADC/DAC to test different stylus designs

- Designed and laid out 10+ PCBs for the touch hardware, including analog front-ends, ADC boards, FPGA boards, and test fixtures
- Checked designs in simulation and on the bench, including HFSS sims of the touch-screen sensing path

AR Modular RF

Seattle, WA

RF Hardware Engineering Intern

Jan 2019 – Apr 2019

- Built LabVIEW/Python automated RF test infrastructure and developed MAC management and firmware flashing automation, significantly improving test throughput and repeatability for IP-enabled RF products
- Performed RF spec analysis and documented technical content for marketing

Evertz Microsystems

Burlington, ON, Canada

Systems Engineering Intern

Sep 2017 – Aug 2018

- Designed DDR3 CPU-to-DMA buffer in RTL for Wireshark-compatible frame capture and developed FPGA RTL pipeline for timestamped 10Gb/s packet capture and replay
- Researched impedance/phase matching for 25G QSFP lanes to improve signal integrity
- Captured RGMII, SGMII, and 1000BTX schematics with optimized magnetics and termination
- Debugged PCB short circuits and engineered active fuse systems

Dozr

Kitchener, ON, Canada

Full-Stack Software Engineer Intern

Jan 2017 – Apr 2017

- Shipped React frontend components with advanced state management and architected a distributed Node.js/Python web crawler with intelligent throttling and parallelism

University of Waterloo

Waterloo, ON, Canada

Undergraduate Research Assistant

Apr 2020 – Aug 2020

- Designed a 60GHz mmWave radar system for non-contact patient vitals monitoring, implementing a wavelet/autocorrelation DSP pipeline achieving reliable breathing rate detection at 10m range
- Created DSP pipeline using wavelets/autocorrelation to detect breathing rates at 10m range
- Built Python/C++ client-server system for real-time offloaded processing

PROJECTS

Most projects are on GitHub; more at mackenzieg.com/projects

Microplastic Detector | *Research Project (2021 – 2022)*

- Designed and fabricated a microwave cavity resonator system achieving reliable detection of microplastic concentrations from 300–900 ppm via resonant frequency shift and S11 analysis
- Developed signal processing algorithms to analyze resonant frequency shifts and S11 magnitude changes

EDUCATION

University of Waterloo

Waterloo, Ontario, Canada

Bachelor of Science in Electrical Engineering

Sep 2016 – June 2022